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RESEARCH ARTICLE

SARMs, Metabolic Modulators and Growth Hormone Secretagogues in Suspected Illegal Medicines, Bought as Sport Performance Enhancers: A Retro- and Prospective Study Within the GEON

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ABSTRACT

Although the abuse of muscle-building compounds in elite sports is already known for a long time, these products have become more popular in recreational sport over the past years. Although anabolic steroids are still the most popular ones in this context, the use of other molecules with anabolic properties is on the rise. Three categories of such products are the selective androgen receptor modulators (SARMs), the metabolic modulators and the growth hormone secretagogues (GHS). Based on this trend and the outcomes of a previous market surveillance study in the domain of illegal products (MSSIP), the Falsified Medicines Working Group of the General European Official Medicines Control Laboratories (OMCL) Network (GEON) decided to conduct an MSSIP with focus on SARMs, metabolic modulators and GHS over a period of 5 years. In total 324 samples and 354 results, reported by 14 laboratories in 13 countries, members of the GEON, were taken into account, for which the majority of the samples originated from illegal distribution. Sixty-five percent of the seized products were represented as medicine, though 24% as dietary supplements, which is of concern because here the (recreational) sporter is not aware he is taking an (unapproved) pharmaceutical. Eighteen different molecules, within the scope of the study, were reported with as top 5: ibutamoren, ligandrol, ostarine, cardarine and andarine. From the limited quantitative data reported,

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it can be assumed that the majority of the samples contain active doses and some are even overdosed, so health risks for the consumers cannot be neglected.

1 | Introduction

Although the phenomenon of doping using muscle-building compounds is already well known in elite sport, its use has found more and more access to recreational sports. This resulted in an extensive illegal market, trading in both doping products, presented as medicines, as well as in a whole series of dietary supplements, possibly adulterated with active pharmaceutical ingredients and so posing as medicines in disguise. The latter is extremely dangerous because in this case, sporters are unaware of taking doping, with the risk of overdose and adverse effects that can not immediately be related to the supplement, as well as risks of interferences with already existing therapies [1, 2]. Typical molecules here are the anabolic agents, with the exogenous steroids still being the most known and popular category. Although anabolic steroids are still popular in this domain, the popularity of other molecules with anabolic properties is increasing. Three categories in the context of this trend are the selective androgen receptor modulators (SARMs), the metabolic modulators and small molecule growth hormone secretagogues (GHS) [3].

SARMs are a category of molecules with similar anabolic properties as anabolic steroids, though with less androgenic effects, and therefore a lot of them are under investigation as medicines for the treatment of, for example, osteoporosis or muscle wasting diseases. Although some of them, more specifically SARMs antagonists like flutamide, nilutamide, bicalutamide and enzalutamide, are now registered as medicine for the treatment of prostate cancer [4], a whole series of molecules is available on the illegal (internet) market [3]. Most of them are still under clinical investigation or even failed clinical studies. All SARMs are prohibited by the WADA because they have the potential to be misused as sport performance enhancement products [5]. Examples here are andarine and ostarine.

Metabolic modulators are a group of molecules that are able to boost muscle metabolism by stimulation of glucose and fatty acid oxidation, mitochondrial biogenesis and muscle remodelling to a fibre type composition. These processes would all improve muscle endurance, and this is without exercise [6]. Typical examples of these kinds of molecules are acadesine and cardarine. Also, here, all metabolic modulators are prohibited by WADA and are not to be used in sports [5].

GHS are a group of molecules with main action the increase of growth hormone release in the human body. They include agonists of the growth hormone secretagogue receptor, for which the natural ligand in the body is ghrelin, and of the growth hormone releasing hormone receptor. GHS were developed to treat or diagnose GH deficiency, resulting in growth retardation, gastrointestinal dysfunction and altered body composition, and some of them are nowadays considered as medicines [7, 8]. Examples are sermorelin, which is FDA approved for the treatment of GH deficiency in children and tesamorelin, which was approved for the reduction of excess abdominal fat in HIV-infected patients with lipodystrophy [7]. Based on the capacity of stimulating the release of growth hormone, it is not a surprise that these molecules can and are used as sport performance enhancers and therefore they are also banned by WADA [5]. An example of a small molecule GHS is ibutamoren.

Besides the use of these products in professional sport settings and so covered by the 'anti-doping' laboratories, their increasing popularity within the general population and their use in recreational sport makes them also a subject for custom laboratories and laboratories dealing with the trade in illegal medicines. These products are generally sold for oral intake in the form of tablets, dietary supplements and liquids, though some websites also sell products for injection. As already mentioned, the majority of these molecules are still under or failed clinical investigations, so the threat to the consumer's health cannot be neglected. Moreover, even if products, registered and recognised as medicine, are used without supervision of a medical doctor, the risks are considerable.

The General European Official Medicines Control Laboratories (OMCL) Network (GEON) is a European network of independent quality control laboratories for medicines and related products, also dealing with the problematics around the trade and use of illegal medicines. The network comprises almost 70 laboratories from more than 40 different member states. These include most of the countries, member of the Council of Europe, but also some non-European ones like Australia, Canada, Israel, Kazakhstan, Morocco, Singapore and Taiwan [9]. Within this network the Falsified Medicines Working group was created with the main objective to share knowledge and technical expertise for the testing of illegal medicines and this by exchanging analytical methods or samples. This group also organises market surveillance studies on suspected illegal products (MSSIP), set up as a retro- and prospective data collection study by the different members over a predefined period of time [10]. In total, five studies were reported with respective subjects: slimming products, sexual potency enhancers, illegal anabolic steroids, medicines in disguise and illegal or illegally traded health products containing non-anatomical therapeutic chemical-international non-proprietary name (non-ATC-INN) molecules [10]. One of the outcomes of the latter study was that molecules falling in the category of the non-steroid anabolic agents gained popularity in the countries participating in the study and that SARMs were part of the top 5 of therapeutic categories in which non-ATC-INN drugs were encountered [11].

Based on these outcomes, the falsified medicines working group decided to conduct a new MSSIP study, focused on illegal or illegally traded SARMs, metabolic modulators and small molecule GHS [11]. The general goal of the study was to get an idea about the products circulating on the European market and beyond, which contain such type of molecules and are illicitly used as physical performance enhancers and to raise awareness of these products and their dangers with both the public as the European and national authorities.

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The study encompassed a retro- and prospective part over a period of 5 years and all OMCL member states were invited to participate. At the end, 14 laboratories from 13 different countries (Australia, Austria, Belgium, Canada, Germany, Italy, the Netherlands, Norway, Poland, Portugal, Spain, Sweden and the United Kingdom) entered the study.

2 | Methods and Materials

2.1 | Scope

The scope of the study was defined as follows: any product (medicine, herbal medicine, dietary supplement, cosmetic, medical device, etc.) containing compounds, such as the following:

- SARMs
- Metabolic modulators
- Small molecule GHS
- Steroidal SARMs (example YK-11)

If quantitative data were available, it was also asked to report them.

Out of scope of the study were products containing anabolic androgenic steroids with the exception of the steroidal SARMs and the GHS of type GHRP-2 and similar substances (illegal peptide drugs).

The samples to be reported were collected by the testing OMCL or the national authorities from either legal or illegal sources, including those obtained in the country of the OMCL via the internet (regardless of the country of origin).

2.2 | Data Collection

All analysis of the suspected illegal products were performed according to the standard operating procedures, available at each individual participating OMCL. Liquid chromatography (LC), hyphenated with (high resolution) mass spectrometry was the most applied technique in the screening step and so the identification of the active component(s). Other techniques were gas chromatography–mass spectrometry, LC-UV and (quantitative) nuclear magnetic resonance analysis. The general approach within the GEON network for the analysis of suspected products, divided in different types of samples, is summarised in the position paper 'An "aide-mémoire" for the testing of suspected illegally traded and falsified medicines' [12].

In the context of quality monitoring throughout the GEON network, the EDQM regularly organises ring trials. For the context of this paper, the ring tests fall under the Suspicious Unknown Products (SUP) programme and consist of a selected unknown sample analysed by all participating OMCLs according to their own procedures. Participation can either confirm the expertise of the OMCL or be the trigger for corrective actions.

The results for the samples within the scope of this study were reported in a harmonised way, using a Microsoft Excel template. The following information was requested:

- Laboratory reference number
- Product name
- Country where the product was sampled
- Sample origin* (commercial shop, internet, law enforcement, etc.)
- Sample source* (legal/illegal)
- Sample provider* (customs, law enforcement, industry, etc.)
- Type of product* (food supplement, cosmetic, falsified medicine, etc.)
- Labelled or suspected purpose of use
- Dosage form*
- List of labelled components/strength
- Name of identified SARMs
- Quantitative results for identified SARMs (optional)
- Measurement uncertainty (optional)
- Approval date of result report
- Analytical method

The features indicated with * were drop-down lists to ensure a harmonised reporting.

2.3 | Timeframe

The study was a combination of both retro- and prospective data and covered the timeframe from January 2018 to the end of September 2023. This time range refers to the date of approval of the results report.

3 | Results

At the end of the reporting time frame, 336 samples corresponding to the scope of the study (Section 2.1) were reported. Four percent of this sample set was excluded (12 samples) because no active compounds, within the scope of this study, could be identified in these samples. Some samples contained more than one of the targeted compounds, which were reported as different results. At the end, a total of 354 results were considered as valid and used in the following discussion.

A complete overview of the data discussed further can be found in Table S1.

3.1 | Origin of the Samples

About 86% of the 324 samples taken into consideration were originating from the illegal supply chain, while for 10% no origin was reported and 4% (13 samples) were considered as

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being sold through a legal distribution chain. The fact that for 10% of the samples, the respective OMCL could not report the origin is often due to a third party seizing the samples, who do not always share information about the sample or on the context of the seizure. The reported origin of the samples is shown in Table 1.

The vast majority of the result set (±76%) were seized and sent to the laboratory by customs or police, which is logical knowing that 86% of all samples originated from illegal distribution. Most of the samples classified under the category 'other' were dietary supplements seized on the legal market by sanitary inspectors. More or less the same situation is reflected by the reported sample providers because 209 of the 356 results (81.5%) were provided by law enforcement (police/customs) and inspectorates of medicines authorities (17.4%). Two of the samples classified as 'other' were samples received from hospitals after the occurrence of adverse reactions.

It should be kept in mind that in the context of this study, samples are also analysed by other laboratories in the respective countries, for example, forensic and customs laboratories. Therefore, the number of samples reported in this study only reflects what is analysed by the OMCLs, and therefore probably many more samples are circulating, seized and analysed throughout the countries of the network.

3.2 | Product Types of Tested Samples

The different types of products encountered in this study are summarised in Table 2. The majority of the samples were presented as a pharmaceutical product, and so we can assume that most consumers were aware of what they were taking because most products had an indication of their real content. Nevertheless, it cannot be assumed that consumers were aware of the fact that for a lot of these products, no clinical or toxicological data are available, or even that some of the molecules, present in these products failed clinical trials due to the occurrence of severe adverse effects. Furthermore, a high number of products were sold as dietary supplements. In this case, at least part of the consumers might be unaware they are taking non-approved pharmaceutical substances, leading to potential side effects, which cannot immediately be related to the supplement nor by the patient nor by health professionals. The five samples, classified as 'other', were labelled as being 'for research purposes only'.

Table S1 also provides an overview of the labelled or suspected indications (based on representation) and the corresponding molecules found for each sample. From this table, it can be seen that in almost half of the cases (49.2%), no indication can be found on the packaging of the samples, while in 43.5% percent of the samples, the (suspected) indication was related to sport performance enhancing. Other indications found on the packaging were 'boost cognitive performance', 'sexual performance enhancer' and 'cardiovascular diseases'. Finally seven samples were sold as 'for research purposes only'.

Table S1 additionally allows a comparison between the labelled ingredients, if any, and the ones found during analysis. In 48% of the cases, active ingredients were labelled on the packaging. From these samples, there was a correspondence between labelled ingredients and the compounds found during analysis in ±88% of the samples. Samples containing the labelled compounds, but also others, were counted as not being corresponding (label vs. content).

3.3 | Dosage Forms of the Tested Samples

Table 3 gives an overview of the different dosage forms encountered during the study. The majority of the samples were presented as solid dosage forms (58%), either capsules, powders or tablets. Forty-one percent of the samples were solutions, either for injection or for oral consumption.

TABLE 1 | Origin of the samples reported by the OMCLs.

Sample origin	Number of test results reported in this category	Percentage
Targeted police/ customs operation	185	52%
Police/customs	84	24%
Internet	36	10%
Unknown	29	8%
Other	10	3%
Commercial shop	9	2.5%
Fitness centre	1	0.3%

TABLE 2 | Sample types reported by the OMCLs.

Product type	Number of results	Percentage
Looking like a pharmaceutical product	233	65%
Food supplements	85	24%
Other	9	3%
Unknown	23	6%
Substance	6	2%

TABLE 3 | Dosage forms of the samples reported by the OMCLs.

Dosage form	Number of results	Percentage
Solution/liquid	146	41%
Capsules	124	35%
Powder	45	13%
Tablets	36	10%
Suspension/ emulsion/paste	5	1%

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3.4 | Detected Molecules

Eighteen different molecules, within the scope of this study, were detected in the different reported samples. Table 4 lists them all, together with the number of times they were reported.

Ibutamoren, also called oratrope, nutrabol or MK-677, was the most reported molecule. Ibutamoren is a small molecule GHS and was identified as a promising treatment for growth hormone deficiency because it mimics the endogenous hormone ghrelin, which stimulates the release of growth hormone. However, the molecule failed clinical trials because several adverse effects were reported, including transient elevation of liver enzymes, hypertension and heart failure. Moreover, only a small increase in muscle mass and bone mineral density could be observed [13, 14].

Ligandrol or LGD-4033 was the second most reported substance and is a SARM, acting as an agonist of the androgen receptor. In preliminary clinical trials, the molecule showed an improvement of lean body mass and muscle strength, though it is still under investigation and therefore not yet approved for human use [15].

Ostarine (enobosarm), cardarine (endurobol) and andarine complete the top 5 of most reported molecules in this study. Ostarine is a SARM under development for the treatment of androgen receptor-positive cancers for its properties to inhibit muscle loss caused by the use of glucagon-like peptide-1 (GLP-1) receptor agonists. Ostarine is also one of the most abused SARMs as a sport enhancer, and cases of liver toxicity were observed with regular users [16]. Cardarine or GW501516 is a metabolic modulator, acting as a selective agonist of the peroxisome proliferator-activated receptor-delta (PPAR-δ) and developed originally for the treatment of metabolic diseases such as obesity and diabetes and cardiovascular diseases. The development was, however, stopped because of serious adverse effects in animal studies [17, 18]. Andarine is another SARM for which research was discontinued and was originally under development as a potential treatment for osteoporosis, benign prostatic hypertrophy and muscle atrophy [18].

TABLE 4 | The detected molecules in the reported samples.

Reported molecule	No. of reports	CAS number
Ibutamoren	62	159634-47-6
Ligandrol	48	1165910-22-4
Ostarine	46	841205-47-8
Cardarine	45	317318-70-0
Andarine	34	401900-40-1
RAD-140	26	1182367-47-0
SR9009	25	1379686-30-2
YK-11	21	1370003-76-1
Meldonium	14	76144-81-5
S-23	10	1010396-29-8
SR9011	6	1379686-29-9
GW-0742	5	317318-84-6
LGD-3303	3	917891-35-1
RU 58841	3	154992-24-2
Acadesine	2	2627-69-2
5α-Hydroxylaxogenin	2	56786-63-1
ACP-105	1	1048998-11-3
AC-262536	1	870888-46-3

RAD-140, also known as testolone or vosilasarm, is also a SARM, which is still under development for the treatment of hormone-sensitive breast cancer [19]. When abused for body building, severe liver injuries were observed [20, 21]. SR9009 (Stenabolic) and the less frequently encountered analogue SR9011, are metabolic modulators that were found to increase the metabolic activity in the skeletal muscle and to reduce obesity in mice [22]. YK-11 or Myostop is a steroidal SARM, acting as a partial agonist of the androgen receptor, which inhibits the protein myostatin. This would lead to muscle growth and better athletic performances [23]. Meldonium, also called mildronate or quaterine, is a metabolic modulator and approved, in some countries, as an anti-ischemia medication to treat cardiovascular diseases. It acts as a fatty acid oxidation inhibitor [24, 25]. S-23 is an investigational SARM, developed as a male contraceptive for its properties to reversibly suppress spermatogenesis [26]. Similarly to cardarine, GW-0742 is a ligand to PPAR-β and δ and shows some promising effects in the treatment of cardiovascular diseases [27]. LGD-3303 is a SARM that showed an influence on sexual preference and would have an additive effect with alendronate in the treatment of osteoporosis [28]. RU 58841 is a non-steroidal anti-androgen that was originally under development as a topical treatment for androgen mediated conditions as acne and alopecia [29, 30]. Acadesine, or shortly AICAR, is an adenosine monophosphate-activated protein kinase activator that is clinically used in the treatment and prevention of cardiac ischemic injury. Today, it is also known as "the exercise pill" [31, 32]. 5α-Hydroxylaxogenin is a synthetic derivative of laxogenin and is a partial agonist of the androgen receptor and sold as dietary supplement for body building [33–35]. ACP-105 is a SARM, originally investigated for its benefits in treating neurological diseases such as Alzheimer's, but is now also abused in sports [36, 37]. Finally, AC-262 is also a SARM and acts as a partial agonist of the androgen receptors with similar effects as ACP-105 and RAD-140 [38].

The following molecules were reported by some of the participants, though they are not discussed because they are out of scope for this study: 2,4-dinitrophenol, caffeine, clomiphene, epimethandienone, letrozole, niacinamide, sildenafil, stanozolol, tadalafil, tamoxifen, testosterone and yohimbine.

It is important to highlight that different combinations of substances were identified. Combinations of two or more sport performance enhancers, e.g., ibutamoren + ostarine + RAD-140, but also combinations with other substances like anabolics, e.g., stanozolol and PDE-5 inhibitors like sildenafil. No one can tell

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here, what the effects, resulting from the consumption of these mixtures can be.

3.5 | Quantitative Results

For almost half of the informed data, the participating OMCLs reported (semi-) quantitative results (Table S1). For 13 of the 18 molecules detected during this study, at least one quantitative result was reported; though, based on the different ways of reporting these results, it is not easy to generate some statistical results, but the results are enough to encourage vigilance. For acadesine, only one sample was found with a dosage of 11 mg/tablet, which corresponds to the dosage often found on websites selling this kind of supplement (10 mg/capsule or tablet). For andarine, websites announce doses of up to 50 mg/mL in liquid form and 25 mg per capsule. Capsule dosage seems to correspond to what is reported, though for the liquids a whole range of concentrations was observed, going from 8 mg/mL to 38.8 mg/g. This seems to be in line with what is sold; at least it is not higher, though also a suspension was reported with a dosage of 43% w/w, which is much higher than the dosage of 50 mg/mL generally sold and therefore this product can be considered very dangerous. Another point of concern here is the seizures of powders with purities between 65% and 100%, probably used for the production of supplements or for self-dosing of the product, e.g., to add to protein shakes. For cardarine, dosages found on internet are 20 mg/mL for liquids and 5–10 mg per capsule. Reported dosages for tablets and capsules were between 2 and 9.8 mg per unit, though for liquids a range between 0.7 and 142.5 mg/g was observed with again unpredictable outcomes when consuming such high-dosed products. For GW0742, only one product was reported with a dosage of 9.6 mg/mL, which seems to correspond to a dosage generally sold on the internet. Also here, powders were detected with purities between 90% and 100%, probably intended for illicit production or self-dosing. Ibutamoren is sold in dosages ranging from 10 to 30 mg per capsule or tablet and under liquid form with dosages around 25 mg/mL. Looking at the reported capsules and tablets, dosages were ranging from 4 to 25 mg per unit, while for the liquids dosages were ranging from 0.4 to 262.6 mg/mL. In fact, two highly dosed liquids were reported, representing a real danger if not diluted before use. Also here, high purity powders were observed in the market. Ligandrol is sold on the internet in the form of capsules and tablets with dosages between 4 and 20 mg, which corresponds perfectly to the reported ranges in this study (4.1–19 mg/unit). Liquids at concentrations of 10 and 20 mg/mL can be purchased; also here matching the reported dosage range (0.6–22.5 mg/mL). High purity powders were reported and could also be purchased on the internet on the same sites as the capsules and liquids, supporting the intended use for self-dosing. The two samples reported containing meldonium had a dosage of around 500 mg per capsule, which is the dosage used as a medicine in the countries where it is recognised for use in the treatment of cardiovascular conditions [24, 25]. Ostarine was found in solid form as highly pure powders and as capsules and tablets with dosages ranging from 0.3 to 48 mg per unit. A quick survey of the internet resulted in products offered at dosages from 10 to 30 mg per unit. Not knowing the background of these samples, it seems that we have both very underdosed and overdosed products circulating. In liquid form, the reported concentration range of 11.4–54.3 mg/mL corresponded to what is offered on the internet (10–50 mg/mL), except for one sample that was found to contain only 1 mg/mL. Next to powders with purities of 24.2% and 94.8%, RAD140 was encountered in solid dosage form of 6–9 mg per unit and liquid dosage forms of 0.2–33.2 mg/g. During the survey on the internet, solid dosage forms and liquids were found in the range of 10–20 mg per unit and 20–40 mg/mL. The very low-dosage liquids are probably fraudulent. So, not only they are sold as illegal products but also they mislead the consumer. S-23 was reported as capsules of about 10 mg per unit and as liquids with concentrations between 3.9 and 26.8 mg/g, which seems to be a normal range circulating on the internet. For both SR9009 and SR9011, several pure powders were encountered, used either for the production of supplements or to be added with a spoon to the diet of the (recreational) sporter. For SR9009, capsules were also analysed with dosages between 6.5 and 11.6 mg/unit as well as liquids ranging from 0.3 to 19.4 mg/mL. For SR9011, liquids with concentrations of 13.7 mg/g and 18.3 mg/mL were encountered during the study. Except for the very low-dosed liquids, the found dosages corresponded to the ones found on vendor's websites, with a range of 5–20 mg per capsule and 10–20 mg/mL for liquids. As for YK-11, high purity powders were detected, as well as two products under capsule form of 0.7 mg and 4.5 mg/unit. In liquid form for oral consumption, the range was 4.1–7.9 mg/mL, and also for one solution, it was indicated to be for injection at a dosage of about 20 mg/mL. Also here, it seemed to correspond to the dosage claimed by the different vendors, though the concentration of 20 mg/mL for injection seems very high.

Although the quantitative results reported by the different OMCLs were difficult to interpret, due to different measurement units and the missing data to recalculate them, it is clear that the majority of the analysed products contained active doses of the compounds and so the side effects described higher can be expected, especially with long-term use. Moreover, some overdosed products can put the patient health at serious risk as well as the highly pure powders, when used for self-dosing and so with an increased risk of overdosing.

4 | Conclusions

From this study, it is clear that illegal products containing SARMs, metabolic modulators and small molecule GHS are circulating in Europe and consumed as sport performance enhancers. This study is based on the reports of 14 OMCLs in 13 different countries, which is quite limited compared with the number of laboratories in the network. Moreover, these are only the results of the OMCLs, often not the only laboratory in the concerned countries performing analysis on the products subject to this study. Forensic and custom laboratories are also analysing. Therefore, it can be assumed that far more products are circulating on the market, covered by the GEON network.

As can be expected for this kind of products, the majority of the samples were originating from the illegal market (86%), while only 4% were seized in what is considered a legal distribution chain.

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Sixty-five percent of the reported samples were presented as a medicine, for which it can be assumed that the consumers bought it, knowing exactly for which purpose and at least have a basic idea of the risks. Though it cannot be assumed that they know they are buying an unapproved drug, which often failed clinical trials, due to severe adverse effects. Also, 24% of the samples was sold as a dietary supplement. Here, the consumer is completely unaware he is taking an (unapproved) pharmaceutical ingredient. These products are called 'medicines in disguise' and are very dangerous because they can cause serious adverse events that will not be immediately related to the product by patient, nor health professional, and they can interfere with already existing treatments.

Some of the substances reported in this study had been identified and investigated as potential medications for a variety of medical conditions. However, for the moment, only meldonium was recognised as a medicine, and a lot of them already failed clinical trials due to severe adverse events. Moreover, the limited quantitative data reported in the study suggest that the majority of the samples contain active amounts of the studied compounds and so adverse effects should be expected. Also different very high-dosed products were reported, representing high risks for the consumers as well as high purity powders, supposedly sold for production of finished dosage forms or for self-dosing. The latter practice also is related to higher risks due to the possibility of overdosing.

In conclusion, this study shows the necessity of continuous efforts in screening and analysing illegal and suspected products for their composition. It not only allows us to gather information about the risks and to have data for court trials but also to investigate new trends in doping products and especially their use in recreational sports. The results of this study are intended to raise awareness among the public and policy makers regarding the dangers posed by these substances and, in more general terms, to raise awareness and discourage the purchase of products (medicines, dietary supplements or life-style products) from suspected websites, concealing their physical address. With this, it can also raise awareness about the fact that the European Union issued a common logo for recognised internet pharmacies [39].

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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